

How Welfare Loss Is Computed

A Teaching Note: Derivation, a Worked Toy Example, and How It
Scales to the Real Chile Model

Working Notes

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1 The Object We’re Computing

Second-order welfare loss relative to the efficient (flex-price) allocation, from Rubbo’s Proposition 3 as extended in this project (full proof in `rubbo_proofs_and_extension.tex`, § “Welfare Function”):

$$\mathbb{W} = \frac{1}{2} \mathbb{E} \left[\underbrace{(\gamma + \varphi) y_t^2}_{\text{output gap}} + \underbrace{\sum_i \kappa_i \pi_{it}^2}_{\text{within-sector}} + \underbrace{\Phi_C(\boldsymbol{\mu}_t, \boldsymbol{\mu}_t) + \sum_s \lambda_s \Phi_s(\boldsymbol{\mu}_t, \boldsymbol{\mu}_t)}_{\text{cross-sector}} \right] \quad (1)$$

where $\kappa_i \equiv \lambda_{D,i} \varepsilon_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ is sector i ’s *welfare weight* — large when the sector is big in the network ($\lambda_{D,i}$) and sticky ($\hat{\delta}_i$ small, so any inflation it does have translates into a lot of costly within-sector price dispersion). $\boldsymbol{\mu}_t$ is the vector of log markup gaps.

Why three terms, not one. A representative-agent closed economy only has the output-gap term. Once you add (i) multiple sectors with sticky prices, each sector generates its own Calvo price-dispersion cost — Term 2; and (ii) non-unit elasticities of substitution across goods, relative-price misallocation across sectors is itself costly even holding each sector’s own dispersion fixed — Term 3. Term 3 is usually the smallest (in the Chile calibration it adds only 3–8% on top of Terms 1+2 — see §4) but it’s not structurally negligible; it’s small here because this model’s production nests are Cobb-Douglas (elasticity exactly 1), which is a property of the functional form, not a generic result.

Because $\mathbb{W}$ is a sum of expectations of *squares* of variables that are themselves linear in exogenous shocks (in a linearized model), $\mathbb{W}$ reduces entirely to a weighted sum of **variances and covariances**. This is the fact that makes every decomposition below exact, not approximate:

$$y_t = \sum_k L_{y,k} \epsilon_{k,t}, \quad \pi_{it} = \sum_k L_{i,k} \epsilon_{k,t} \quad (2)$$

for structural shocks $\epsilon_{k,t}$ with $\text{Cov}(\epsilon_k, \epsilon_{k'}) = 0$ for $k \neq k'$ (the model’s shocks are mutually independent by construction). Then $\text{Var}(\pi_i) = \sum_k L_{i,k}^2 \sigma_k^2$ — additive across shocks, with no cross-terms, *precisely because independence kills the covariance terms*. This is the whole trick behind “shock k contributes $X\%$ of the welfare loss.”

2 Full Derivation: From Utility to Welfare Loss

Eq. (1) isn't an assumption — it's what a second-order Taylor expansion of household utility around the efficient allocation produces, once you account for Calvo price stickiness and CES/Cobb-Douglas aggregation. This section derives each of the three terms in turn, at a level of detail one notch below `rubbo_proofs_and_extension.tex`'s "Proposition 3" proof outline (§6 there) — the goal here is to make every algebraic step explicit rather than just stating the result.

2.1 Term 1: the output-gap term, from household utility directly

Household period utility is $U(C_t, L_t) = u(C_t) - v(L_t)$ with $u(C) = \frac{C^{1-\gamma}}{1-\gamma}$, $v(L) = \frac{L^{1+\varphi}}{1+\varphi}$. Let (C_t^*, L_t^*) denote the efficient (flex-price, correctly-subsidized) allocation, and write $\tilde{c}_t = \log(C_t/C_t^*)$, $\tilde{l}_t = \log(L_t/L_t^*)$ for log-deviations from it.

Expand $u(C_t)$ to second order. Writing $C_t = C_t^* e^{\tilde{c}_t}$ and Taylor-expanding the exponential to second order in $\tilde{c}_t$:

$$u(C_t) - u(C_t^*) = \frac{(C_t^*)^{1-\gamma}}{1-\gamma} \left[(1-\gamma)\tilde{c}_t + \frac{1}{2}(1-\gamma)^2\tilde{c}_t^2 \right] + O(\tilde{c}_t^3) = U_C^* C_t^* \left[\tilde{c}_t + \frac{1}{2}(1-\gamma)\tilde{c}_t^2 \right] \quad (3)$$

using $U_C^* \equiv u'(C_t^*) = (C_t^*)^{-\gamma}$, so $U_C^* C_t^* = (C_t^*)^{1-\gamma}$. The symmetric calculation for $-v(L_t)$ (with $V_L^* \equiv v'(L_t^*) = (L_t^*)^\varphi$) gives

$$-v(L_t) + v(L_t^*) = -V_L^* L_t^* \left[\tilde{l}_t + \frac{1}{2}(1+\varphi)\tilde{l}_t^2 \right] \quad (4)$$

Adding them:

$$\Delta U_t \equiv U_t - U_t^* = U_C^* C_t^* \tilde{c}_t - V_L^* L_t^* \tilde{l}_t + \frac{1}{2} U_C^* C_t^* (1-\gamma)\tilde{c}_t^2 - \frac{1}{2} V_L^* L_t^* (1+\varphi)\tilde{l}_t^2 \quad (5)$$

Killing the linear terms. Two facts collapse the first (linear) bracket to exactly zero:

- (i) **Goods-market clearing to first order:** $\tilde{c}_t = \tilde{y}_t$.
- (ii) **Lemma 2 (output gap = employment gap, `rubbo_proofs_and_extension.tex` §4):** $\tilde{y}_t = \tilde{l}_t$, because sticky prices only distort the *aggregate* labor margin — the

allocation of a given aggregate L_t across sectors stays efficient since relative wages are flexible.

So $\tilde{c}_t = \tilde{l}_t = \tilde{y}_t$ *exactly* at this order. Combined with the efficient labor-supply condition $U_C^* C_t^* = V_L^* L_t^*$ (the standard welfare-theorem normalization: at the efficient point, the household's MRS equals the marginal product, which after the usual optimal employment subsidy correcting the monopoly markup makes the two curvature-weighted levels equal), the linear term is

$$U_C^* C_t^* \tilde{c}_t - V_L^* L_t^* \tilde{l}_t = U_C^* C_t^* (\tilde{y}_t - \tilde{y}_t) = 0 \quad (6)$$

exactly — not approximately, because $\tilde{c}_t$ and $\tilde{l}_t$ are the *same* variable here, not merely correlated.

What's left is purely the quadratic terms in Eq. (5), all in $\tilde{y}_t$ (via $\tilde{c}_t = \tilde{l}_t = \tilde{y}_t$) and all sharing the common factor $U_C^* C_t^*$:

$$\Delta U_t = \frac{1}{2} U_C^* C_t^* \left[(1 - \gamma) - (1 + \varphi) \right] \tilde{y}_t^2 = -\frac{1}{2} (\gamma + \varphi) U_C^* C_t^* \tilde{y}_t^2 \quad (7)$$

Dividing through by $U_C^* C_t^*$ (the standard normalization to express welfare loss in consumption-equivalent units) and flipping sign to define $\mathbb{W} \equiv -\Delta U_t / U_C^* C_t^* \geq 0$ gives exactly Term 1 of Eq. (1): $\frac{1}{2} (\gamma + \varphi) \tilde{y}_t^2$.

The output-gap term needed *no* information about the production network, Calvo parameters, or number of sectors — it comes purely from curvature of preferences (γ, φ) and the two identities $\tilde{c}_t = \tilde{y}_t = \tilde{l}_t$. All network/price-stickiness structure enters only through Terms 2 and 3, and indirectly, through how $\tilde{y}_t$ itself is determined in equilibrium (the Phillips curve system).

2.2 Term 2: within-sector price dispersion, from Calvo pricing

Within sector i , firms that reset at different past dates charge different prices $P_{if,t}$, even though they face identical current marginal cost. This dispersion is directly costly: producing a given amount of the sector- i CES aggregate $Y_{it} = \left(\int_0^1 Y_{if,t}^{(\varepsilon_i-1)/\varepsilon_i} df \right)^{\varepsilon_i/(\varepsilon_i-1)}$ from dispersed-price varieties wastes labor/inputs relative to charging one uniform price, because demand $Y_{if} = (P_{if}/P_i)^{-\varepsilon_i} Y_i$ shifts disproportionately toward the (arbitrarily) cheaper varieties without them being any more efficient to produce.

The dispersion wedge. Define $\Delta_{it} \equiv \int_0^1 (P_{if,t}/P_{it})^{-\varepsilon_i} df \geq 1$ (equality iff all firms charge the same price). A second-order expansion of Δ_{it} in the cross-sectional log-price distribution gives the standard result (Yun 1996; Woodford 2003, ch. 6):

$$\log \Delta_{it} \approx \frac{1}{2} \varepsilon_i \text{Var}_f[\log P_{if,t}] \quad (8)$$

— a pure Jensen’s-inequality cost of the CES aggregator, increasing in both the elasticity ε_i (more substitutable varieties $\Rightarrow$ demand reallocates more aggressively toward mispriced ones) and the cross-sectional price variance.

Mapping dispersion to inflation. Under Calvo, $\text{Var}_f[\log P_{if,t}]$ is not a free parameter — it’s pinned down by the sector’s own inflation history, because dispersion only arises from *staggering*: firms that reset at t all charge the identical P_{it}^* , so all cross-sectional variance comes from the gap between resetters and the (persisting) distribution of past prices. Log-linearizing the recursion $\text{Var}_f[\log P_{if,t}] = (1 - \delta_i)(\text{Var}_f[\log P_{if,t-1}] + (\text{new dispersion from this period’s inflation}))$ around the zero-inflation steady state and solving forward (Yun 1996; the algebra is a standard but lengthy application of the Calvo recursion, omitted here) yields

$$\text{Var}_f[\log P_{if,t}] \approx \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \pi_{it}^2 \quad (9)$$

with $\hat{\delta}_i$ the same effective Calvo parameter defined in `rubbo_proofs_and_extension.tex` Eq. (19) (“Calvo Optimal Reset Price”). Intuitively: a stickier sector ($\hat{\delta}_i$ small) needs a *larger* cross-sectional variance to generate the same observed aggregate inflation π_{it} , because fewer firms are moving to produce that inflation number, so the ones that do move must move further apart from the frozen majority.

From dispersion to a TFP-equivalent utility loss. $\log \Delta_{it}$ acts exactly like a negative TFP shock to sector i : producing sector output Y_{it} requires $\Delta_{it} \times$ (what would be needed absent dispersion) units of underlying labor/inputs. Its contribution to aggregate consumption-equivalent loss is this TFP loss weighted by the sector’s **Domar weight** $\lambda_{D,i}$ (its total-sales share, counting indirect sales through the network) — the same weighting Lemma 1 uses to build *natural* output from sectoral TFP:

$$\text{Term 2} = \sum_i \lambda_{D,i} \varepsilon_i \cdot \frac{1}{2} \text{Var}_f[\log P_{if,t}] \stackrel{(9)}{=} \frac{1}{2} \sum_i \underbrace{\lambda_{D,i} \varepsilon_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}}_{\kappa_i} \pi_{it}^2 \quad (10)$$

exactly Term 2 of Eq. (1).

2.3 Term 3: cross-sector misallocation, from non-unit substitution elasticities

Terms 1–2 together would be the *entire* welfare loss if every aggregator in the model (household consumption bundle, each firm’s input bundle) were Cobb-Douglas — because Cobb-Douglas has unit elasticity of substitution, and a firm/household with unit elasticity is, to second order, *indifferent* to small relative-price distortions across its inputs (it substitutes away from the relatively-expensive input exactly enough to keep expenditure shares fixed, which is precisely what makes the resulting misallocation cost vanish at second order). Term 3 exists only because this model’s *outer* nest — $C_t = \text{CES}(C_t^H, C_t^F; \eta)$, domestic vs. imported consumption — has $\eta = 1.5 \neq 1$.

General second-order misallocation cost (Baqae–Farhi). For any CES-type aggregator $G(X_1, \dots, X_n)$ with cost shares w_k and pairwise Allen–Uzawa elasticities σ_{kh} , a vector of small *wedges* x_k (log deviations of X_k from its cost-minimizing/efficient level) generates a second-order TFP-equivalent loss

$$\Phi_G(\mathbf{x}, \mathbf{x}) = \frac{1}{2} \sum_{k,h} w_k w_h \sigma_{kh} (x_k - x_h)^2 \quad (11)$$

— a Harberger-triangle-style formula: quadratic in the wedge *differences* (only relative distortions matter, not the level), scaled by how substitutable the pair is. This is the general form; deriving it from a second-order expansion of G itself is standard in the production-network literature (Baqae & Farhi 2019, 2020) and not reproduced here.

The wedges here are markup gaps. In this model, the object playing the role of x_k is the log markup gap $\mu_{it} = \log(P_{it}/MC_{it}) - \log \frac{\varepsilon}{\varepsilon-1}$ — the deviation of sector i ’s markup from its flexible-price target — applied once at the household’s consumption nest (Φ_C , weights β^H) and once per firm’s production nest (Φ_s , weights $(\alpha_s, \Omega_s^H, \Omega_s^F)$), summed with Domar weights on the latter:

$$\text{Term 3} = \Phi_C(\boldsymbol{\mu}_t, \boldsymbol{\mu}_t) + \sum_s \lambda_{D,s} \Phi_s(\boldsymbol{\mu}_t, \boldsymbol{\mu}_t) \quad (12)$$

exactly Term 3 of Eq. (1). The closed-form collapse to a *weighted variance* of $\boldsymbol{\mu}_t$ used in §4 (and in `code/cross_sector_welfare.py`) follows from Eq. (11) whenever $\sigma_{kh} \equiv \sigma$ is

constant across all pairs — true here since $\sigma_{kh} = 1$ (Cobb-Douglas) for every domestic-domestic and domestic-labor-import pair, and only the domestic-vs.-import-composite pairs get $\sigma_{kh} = \eta$.

Why the three terms have the shape they do. Term 1 comes purely from preference curvature and holds with *no* reference to price stickiness at all (it would exist even with flexible prices, if output happened to deviate from efficient for some other reason). Terms 2 and 3 both come from the *same* underlying source — CES aggregation of heterogeneous prices — but at two different levels of the model: Term 2 is dispersion *within* a sector’s own variety continuum (entirely a Calvo/stickiness artifact, vanishes if $\delta_i = 1$), while Term 3 is misallocation *across* sectors/goods in a bundle (a property of the aggregator’s curvature σ_{kh} , vanishes only if every relevant elasticity is exactly 1, regardless of stickiness). This is also why Term 3 needed a genuinely new elasticity object (η) that Terms 1–2 never used.

3 Worked Toy Example

To make this concrete before touching the real 3-sector Chile model, consider a minimal toy economy with **two** sectors: Sector 1 (upstream, directly FX-exposed) and Sector 2 (downstream, buys from Sector 1, resets prices slowly). Three independent shocks hit each period: χ_1, χ_2 (sector TFP) and Δe (FX depreciation). Full code: `toy_welfare.py` (scratchpad).

3.1 Primitives

	Sector 1 (upstream)	Sector 2 (downstream)
Domar weight λ_i	0.30	0.70
Calvo reset prob. $\hat{\delta}_i$	0.40	0.15
Welfare weight $\kappa_i = \lambda_i \varepsilon^{\frac{1-\hat{\delta}_i}{\hat{\delta}_i}}$	2.70	23.80

($\varepsilon = 6$, $\gamma + \varphi = 1$ for both sectors — Sector 2’s κ is nearly $9\times$ Sector 1’s despite being only $2.3\times$ larger in Domar weight, because it is also far stickier.)

3.2 Impact loadings (a stand-in for the full Phillips-curve solution)

Instead of solving a full DSGE, assign each variable a linear loading on each shock directly — this is exactly what the real model’s $\mathcal{B}, \mathcal{V}, \Gamma$ vectors do (network-amplified loadings), just without deriving them from IO data:

	χ_1	χ_2	Δe
π_1	1.00	0.00	0.25
π_2	0.40	1.00	0.08
y	0.50	0.50	−0.20
μ_1	0.50	0.00	0.12
μ_2	0.20	0.50	0.04

Sector 2’s loading of 0.40 on χ_1 (a shock to Sector 1) is the network channel: Sector 2 inherits cost-push from its upstream supplier even though χ_1 never touches it directly — the same logic as $\mathcal{V} = \Delta(I - \Omega\Delta)^{-1}(\cdots)$ in the real model. Shock sizes: $\sigma_{\chi_1} = \sigma_{\chi_2} = 0.010$, $\sigma_{\Delta e} = 0.020$ (FX more volatile, as calibrated in the real model).

3.3 Results

Term decomposition (Eq. 1, evaluated numerically):

Term	Value ($\times 10^{-3}$)	Share
Output-gap	0.033	2.0%
Within-sector	1.580	97.7%
Cross-sector	0.0038	0.2%
Total W	1.616	100%

Within-sector dispersion dominates — same qualitative pattern as the real Chile result (§4), and for the same structural reason: Cobb-Douglas cross-sector nests make the cross-sector term small *by construction* (elasticity exactly 1), not because misallocation is empirically unimportant.

Sector decomposition of the within-sector term:

	Contribution ($\times 10^{-3}$)	Share of within-sector
Sector 1 (upstream)	0.169	10.7%
Sector 2 (downstream/sticky)	1.411	89.3%

Despite Sector 1 having *larger* exposure to the FX shock directly, Sector 2 absorbs almost 90% of the within-sector loss — because $\kappa_2 \gg \kappa_1$ dominates $\text{Var}(\pi_1) > \text{Var}(\pi_2)$. This is the toy version of the real model’s “Why Services?” result: size $\times$ stickiness beats direct exposure.

Shock decomposition of total $\mathbb{W}$ (exact, by additivity of Eq. 2):

Shock	Contribution ($\times 10^{-3}$)	Share
χ_1 (Sector-1 TFP)	0.339	21.0%
χ_2 (Sector-2 TFP)	1.206	74.6%
Δe (FX depreciation)	0.073	4.5%

Even though the FX shock is *twice* as volatile as either TFP shock, it contributes the least to welfare loss — because neither sector’s κ_i nor its Γ_i -type loading on Δe is large enough to compound with that extra volatility. This is exactly the kind of result that looks surprising until you decompose it, and exactly why the real project’s risk-premium-volatility sweep (§4.3) was necessary rather than assumed.

Between/within split of the within-sector term. Term 2 itself further decomposes, exactly like weighted ANOVA, into an *aggregate*-inflation piece and a *dispersion*-around-that- aggregate piece:

$$\sum_i \kappa_i \pi_{it}^2 = \kappa_{\text{tot}} \bar{\pi}_t^2 + \kappa_{\text{tot}} \sum_i w_i (\pi_{it} - \bar{\pi}_t)^2, \quad w_i \equiv \kappa_i / \kappa_{\text{tot}}, \quad \bar{\pi}_t \equiv \sum_i w_i \pi_{it} \quad (13)$$

an identity ($E[X^2] = (E[X])^2 + \text{Var}(X)$ applied with κ -weights), true state-by-state, not just on average. In the toy example:

	Value ($\times 10^{-3}$, raw, no $\frac{1}{2}$)	Share of Term 2
Between ($\kappa_{\text{tot}} \bar{\pi}^2$)	2.801	88.7%
Within (dispersion around $\bar{\pi}$)	0.358	11.3%

Most of the “within-sector” term is, in this toy, actually the level of a κ -weighted average inflation rate moving, not sectors diverging from one another. **Caveat:** $\bar{\pi}_t$ here uses the *welfare* weights κ_i (Domar $\times$ rigidity), not the consumption weights β^H

that define the true CPI — so $\bar{\pi}_t$ is a welfare-relevant proxy, not literally π_t^C . The two coincide only if $\kappa_i \propto \beta_i^H$, which need not hold since κ_i also carries the rigidity ratio $(1 - \hat{\delta}_i)/\hat{\delta}_i$.

Regime decomposition (toggling how much of the FX shock passes through, as a stand-in for Float/Managed/Peg):

Regime	Output-gap	Within-sector	Cross-sector	Total
Float (full pass-through)	0.033	1.580	0.0038	1.616
Managed (50% offset)	0.027	1.530	0.0036	1.560
Peg (0% pass-through)	0.025	1.520	0.0036	1.545

Why this toy says “Peg is best,” while the real project says “Peg is worst.” In this toy, killing FX pass-through only ever *removes* a cost channel — there’s no offsetting cost to pegging. The real model’s Peg is dominated instead because pegging doesn’t eliminate the risk-premium/UIP shock, it just closes off the one instrument (the exchange rate) that could otherwise absorb it, so that shock’s variance gets pushed into π_i and y instead. This toy doesn’t have a risk-premium shock at all, which is exactly why it can’t reproduce that result — a useful illustration that the shock decomposition step is not optional set-dressing, it’s what determines the regime ranking. See §4.3.

4 Scaling to the Real Chile Model

The real pipeline is structurally identical to the toy above; only the loadings $L_{i,k}$ come from solving the full linearized DSGE in Dynare (via the QZ decomposition, `pol_fct.m`-style) instead of being assigned by hand, and there is a genuine third sector plus a fourth shock type (monetary policy).

4.1 Where each term is computed

Term	File	Object
Output-gap + within-sector	<code>welf.m</code> / Dynare <code>oo_.var</code>	$50*((\text{gamma}+\text{phi})*y^2 + \text{pi})$
Cross-sector	<code>code/cross_sector_welfare.py</code>	$\Phi_C + \sum_s \lambda_s \Phi_s$, closed form (
Shock decomposition	<code>results/variance_decomposition.csv</code>	Dynare’s native <code>oo_.varian</code>

`cross_sector_welfare.py` reads the full 3×3 unconditional covariance matrix of the markup gap μ_t (`results/markupgap_chile_covar.csv`, from reporting variables MARKUPGAP1/2/3 added to the `.mod` files) rather than just each sector’s own variance — necessary because Φ_C, Φ_s are quadratic forms in the *full* covariance matrix, exactly as in the toy’s $\text{Cov}(\mu_1, \mu_2)$ step.

4.2 Headline numbers (term decomposition, real Chile calibration)

$(\times 10^{-4})$	Float	Managed	Peg
Output-gap + within-sector	25.47	10.17	102.05
Cross-sector (added 2026-07-23)	0.84	0.86	4.57
Total	26.31	11.03	106.62

Same qualitative pattern as the toy: cross-sector is small relative to within-sector (3–8% here vs. 0.2% in the toy — the real model’s outer CES nest, $\eta = 1.5 \neq 1$, makes it somewhat larger than the toy’s all-Cobb-Douglas case, but still second-order).

4.3 Shock decomposition (real model)

`results/variance_decomposition.csv` is Dynare’s own forecast-error variance decomposition, computed the same additive way as §2’s Step 4 (independence of shocks). Reading it for `y_gap`: under Float, the risk-premium shock `eps_rp` explains 68.9% of output-gap variance; under Peg, 85.8%; under Managed, 22.2%. This is the real-model analogue of the toy’s Step 4 table — and it is precisely because `eps_rp` dominates so heavily under Peg (not eliminated by pegging, since UIP/risk-premium is a financial-account object, not an exchange-rate object) that Peg ends up *worst*, the opposite of the toy’s naive “less FX exposure is always better” intuition. Confirmed directly in the risk-premium-volatility sweep (`code/sweep_risk_premium_chile.m`): switching `eps_rp` off entirely makes Peg *roughly tied with Float*, fractionally better — see `results_summary.md`.

4.4 Sector decomposition (real model)

Mirrors the toy’s Step 3 exactly: Domar weight $\lambda_{D,i}$ (size) times a rigidity multiplier $(1 - \hat{\delta}_i)/\hat{\delta}_i$ (stickiness) gives κ_i ; combined with each sector’s simulated $\text{Var}(\pi_i)$, Services (the most Domar-weighted, stickiest sector in the Chile calibration) absorbs the largest share of loss in every regime — the “Why Services?” decomposition in `code/services_mechanism_decomposition.py`, full writeup in `results_summary.md`.

The four decompositions — term, sector, shock, regime — are not four different techniques. They are the *same* additive-variance identity (Eq. 2) sliced along different axes: sum over shocks k (term/sector decomposition, holding the object fixed) or sum over objects (shock decomposition, holding the shock fixed), or recompute the whole object with one loading zeroed out (regime decomposition). Once you have the loadings $L_{i,k}$ and shock variances σ_k^2 , every one of these tables is a different reduction of the same matrix $L_{i,k}^2 \sigma_k^2$ — nothing more is needed.